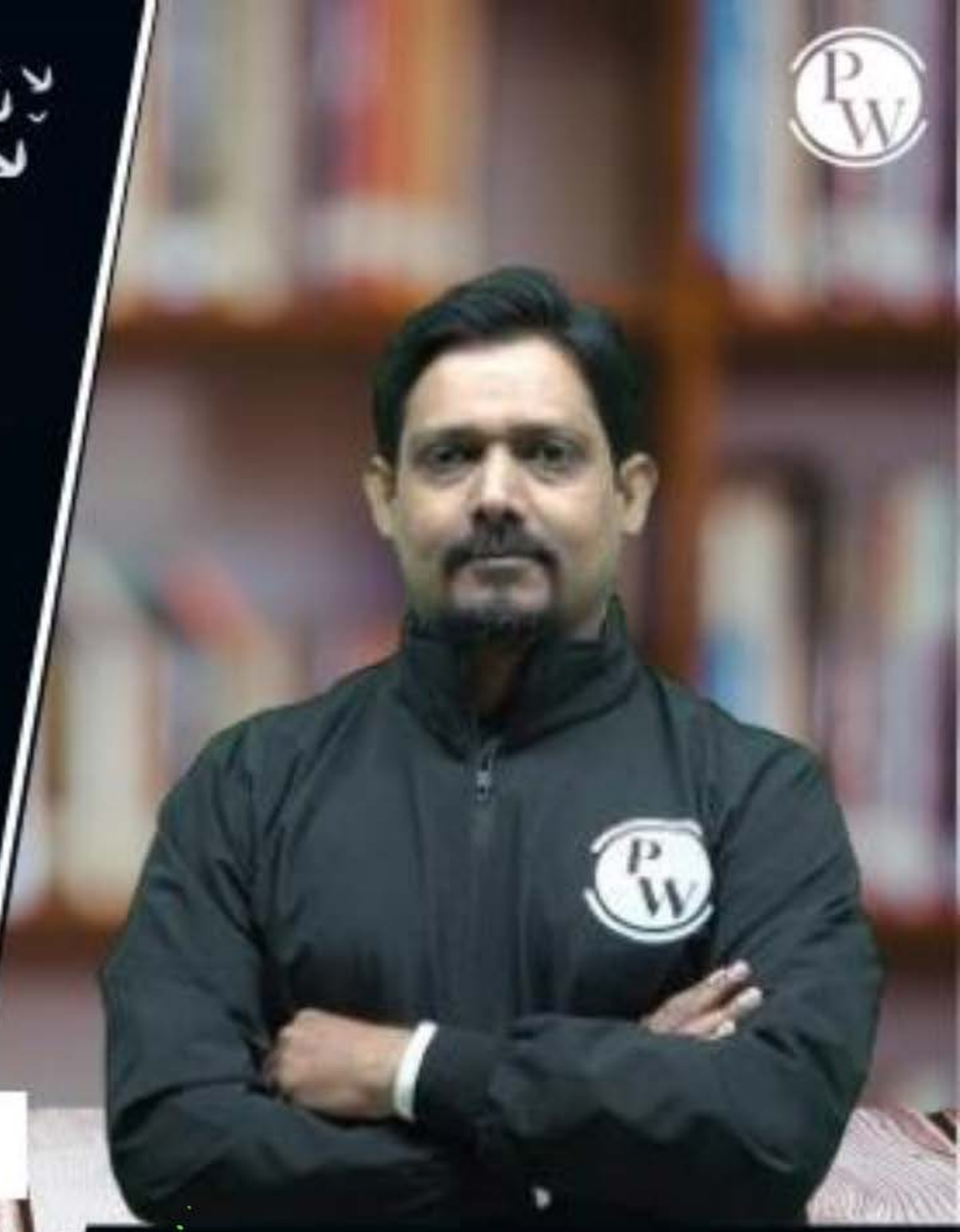
Computer Science & IT



Data Structure & Programming

Stack

Lecture No. 02



By- Abhishek Sir

Recap of Previous Lecture



Topic	Stack as ADT
Topic	Implementation of Stack
Topic	Practice Problem
Topic	
Topic	

Application of Stack

Slide

Topics to be Covered



- Topic — Stack as Permutation Generator
- Topic — Expression In computer
- Topic — Practice Problem
- Topic
- Topic

Slide



Topic : Stack as Permutation Generator

Consider 3 character that will come in order 'a', 'b', 'c'
Character pushed in stack in given order but character can be poped & print anytime.

"Using above property we can generate different permutation of stack"

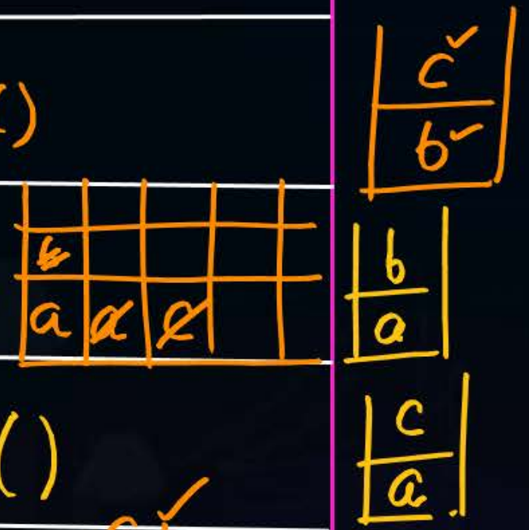
Slide



Topic : Stack as Permutation Generator

5 permutation possible

abc	push(a) pop() push(b) pop() push(c) pop() ✓
acb	push(a) pop() push(b) push(c) pop() pop()
bac	push(a) push(b) pop() pop() push(c) pop
bca	push(a) push(b) pop() push(c) pop() pop()
✗ cab	push(a) push(b) push(c) pop() pop() pop()
cba	push(a) push(b) push(c) pop() pop() pop()



Slide

Topic : Stack as Permutation Generator

n = 4 (5) (12) 5+5+3+1 = (14) 14

push(a)
pop



abcd	bacd 5	cabd ✗	dabc ✗	
abdc	badc ✓	cadb ✗	dacb ✗	
acbd	bcad ✓	cbad ✓	dbac ✗	
acdb	bcda ✓	cbda ✓	dbca ✗	
adbc ✗	bdac ✗	cdab ✗	dcab ✗	
adcb	bdca ✓	cdba ✓ (3)	dcba ✓	

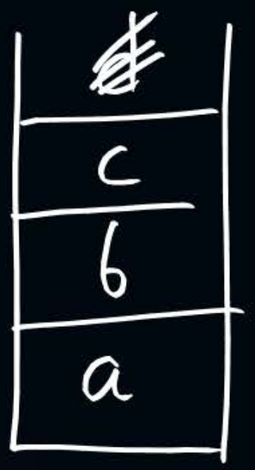
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Topic : Stack as Permutation Generator

- first character d

push(a) push(b) push(c) push(d) pop()





Slide

Topic : Stack as Permutation Generator



b a c d push(a) push(b) pop() pop() push(c) pop() push(d) pop()

b a d c push(a) push(b) pop() pop() push(c) push(d) pop() pop()

b c a d ——,—— pop() push(c) pop() pop() push(d) pop

b c d a ——,—— pop() push(c) pop() push(d) pop() pop()

b d a c ——,—— pop() push(c) push(d) pop() ✗ ✗

b d c a ——,—— pop push(c) push(d) pop() pop() pop()







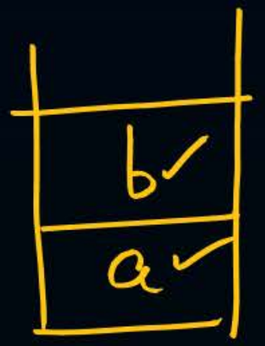


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Topic : Stack as Permutation Generator

push(a) push(b) push(c) pop



Slide



Topic : Stack as Permutation Generator

$n_{C_r} = \frac{\lfloor n}{\lfloor n-r \; \lfloor r}$

$C_1 = 1$ $C_6 = 132$ n distinct character

$C_2 = 2$

$C_3 = 5$

$C_4 = 14$

$C_5 = 42$

No. of permutation : $\frac{1}{n+1} {}^{2n}C_n$ Catalan No.

$C_4 = \frac{1}{5} {}^{8}C_4 = \frac{1}{5} \frac{8 \times 7 \times 6 \times 5}{4 \times 3 \times 2 \times 1} = 14$: ${}^{8}C_4 = \frac{\lfloor 8}{\lfloor 4 \; \lfloor 4}$

$= \frac{8 \times 7 \times 6 \times 5 \; \lfloor 4}{4 \times 3 \times 2 \times 1 \; \lfloor 4}$

$C_5 = \frac{1}{6} {}^{10}C_5 = \frac{1}{6} \frac{10 \times 9 \times 8 \times 7 \times 6}{5 \times 4 \times 3 \times 2 \times 1} = 42$

Slide

Topic : GATE 1994

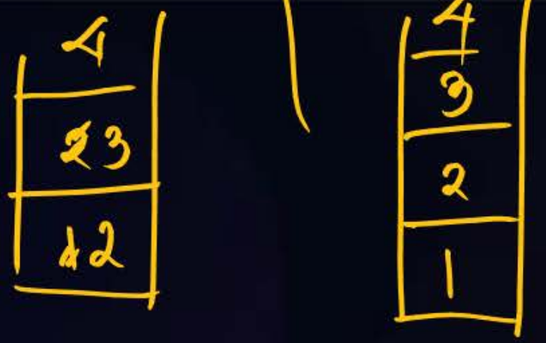
Which of the following permutations can be obtained in the output (in the same order) using a stack assuming that the input is the sequence $1, 2, 3, 4, 5$ in that order?

A. $3, 4, 5, 1, 2$

B. $3, 4, 5, 2, 1$ – B

C. $1, 5, 2, 3, 4$

D. $5, 4, 3, 1, 2$



Slide



Topic : GATE 2004

A program attempts to generate as many permutations as possible of the string 'abcd' by pushing the characters a, b, c, d in the same order onto a stack, but it may pop off the top character at any time. Which one of the following strings CANNOT be generated using this program?

(A) abcd ✓

(b) dcba ✓

(c) cbad ✓

(d) cabd ✓





Slide



Topic : Expression In Computer

a+b×c/d ← Infix Expression

Expression evaluation bound to two Rules

* precedence of operator decides order of evaluation of expression
* if precedence is same then order of evaluation decided by Associativity Rule.

Slide



Topic : Expression In Computer

-5+7

= 2

( )	Bracket	Highest
-(unary)		
$, ↑ or ^ ← Not a C Language symbol	power, Exponentiation	Right to left 2↑3↑2 : 2↑9 : 512
/, *	Multiplication/Division	Left to Right 6/2*4 : 3*4 : 12
+, -	Addition or Substraction	Left to Right 5-4+3 : 1+3 : 4
=	Equal	Right to left (lowest)

Slide



Topic : Expression In Computer

#Q 2016 The attribute of 3 operators are given

Operator	precedere	Associativity
+	High	Left to Right
−	Medium	Right to left
*	Low	Left to Right

The value of expression 2-5+1-7*3 is _____?

Slide



Topic : Expression In Computer

$2 - \underline{5 + 1} - 7 * 3$

$2 - (6 - 7) * 3$

$2 - (-1) * 3$

$3 * 3 = 9$

Slide



Topic : Expression In Computer

a+b ⟶ Infix

+ : operator

a, b or Any Numeric value

Operand

prefix : + a b

operator operand$_1$ operand$_2$

postfix : a b +

operand$_1$ operand$_2$ operator

Slide



2 mins Summary

Topic

permutation using stack

Topic

Expression in computer

Topic

Topic

Topic

Topic

Slide

THANK - YOU